
IOT in cisco packet tracer (fire detection in cisco packet)

Part1

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Abstract

The Internet of Things or the Internet of Things is a system of connected computing devices, mechanical and digital machines, objects, animals, or people with sensors, software, and other technologies to connect and exchange data with other devices and systems via the Internet. are embedded with unique identifiers (UIDs) and the ability to transfer data over a network without the need for human-to-human or human-computer interaction. Fire Detectors play a very important role in Industries, Shops, Malls, Residential complexes, and parking areas. They help in detecting fire or smoke at an early stage and can help in saving lives. Commercial Fire detecting systems usually have an alarm signaling, with the help of a buzzer or Siren. We have designed an IOT based Fire Alerting System using a Temperature sensor, flame sensor, and smoke sensor in cisco packet. This project would not only signal the presence of fire in a particular premise but will also send related information through IOT.

1. Introduction

We will discuss the new topic regarding the iot. in the iot we are going to design a fire monitoring system. If the fire will be detected, the water sprinkler sensor get starting and it spreads water in that area. So this is our scenario, what it will do, what the fire monitoring system will do if you will bring the fire. In the air to the fire detector, it will check the fire and little signal of high or low will be sent based on the current detection. In this here, high signal for fire and low signal for no fire detected, so it will send two signal (high & low). What is a fire detector, definition

A fire detector is a device that detects the presence of a fire. This can be done in several ways, including the following:

Smoke detectors use an optical or electrochemical sensor to detect smoke.

Combustible gas detectors use an electrochemical cell or catalytic sensor to detect combustible gases in the air.

Ionization smoke detectors are capable of detecting large and small particles in the air, as well as large amounts of smoke. Disused due to the use of radioactive components.

Features:

- Registration Server Compatible
- Detects flames by checking for a property and finding if the "IR" property value is in the range the detector considers a fire and outputs a digital signal.

Usage:

- Objects that can be on fire should have a device property "IR" with value. The IR range that is detected as fire is specified in the script. Anything near the detector end will be checked for fire and a digital signal of HIGH or LOW will be sent based on the current detection. HIGH for fire, LOW for no fire detected.

Data Specifications:

- Message Format: [state]
- state: 0 = no fire, 1 = fire
- Output Slot: D0
- LOW = no fire, High = fire

Example:

- Create an Thing and in the script set a fire property with `setDeviceProperty(getName(), 'IR', 900);`
- Connect the fire sensor to a LED that has been modified to check for a digital signal, and turn the LED on when the signal is not 0.
- Move the detector near the fire to see the LED light up.

Fire Sprinkler:

A Sprinkler that puts out fire.

Features:

- Registration Server Compatible
- Raises the water level

Data Specifications:

- Message Format: [state]
- state: 0 = off, 1 = on

To begin, we open new project in cisco packet and in this project we need:

1-fire sprinkler

2-microcontrollers (MCU-PT) board

3-fire monitor

4-heating element

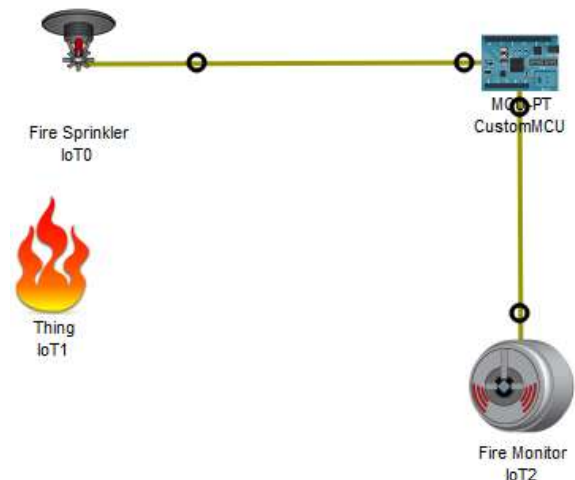


Figure1. It is a part of home automation (fire detection) in Cisco Packet.

We start the work, with take this cable connect with the D(0) here and also connect D(0) here this is my input pin, if you will see D(0) is my input pin so it will take input temperature. you will take the input from the fire monitoring, so if the fire is detected then it will give the out put from the other pin, so we will select the pin here D(1), D(1) will be

the out put it will produce out put on D(1), now it is connected.

So what we need to do now I will write a sample program in MCU. We will click it here and so in the program so click on the program language and you can see the program here we need to selected the python. We write a new program and create new name (firedetector). And you need to write a start program. So let me write the program here first we need to (import*).

```
from gpio import *
from time import *
```

So first what we need to do, please let me know tell me new from (GPIO) pin, you need to import this file, so you need to import this library. So then you need to also import (time library) from time import*. So after that what we need to do we need to write a function, so for function writing, just writing (def) and this the function name (handlesensordata) so this is function name it will.

So first I thought I can write a program then I will explain you line by line, so no I'm going to explain you line by line because the program is almost ready. So what to do now I will explain you have, so we set a function called (handlesensordata) this is we set this function, so what it will read a (value by using this function). It will create detail read this is again one more function we called it here digital read function it will read a value from port number zero, because it will read this MCU will read the value from port number zero. So this function will read the value and save it in this variable value we just named it value, this is variable, so whatever the value it will read it save it in this variable.

If the value is zero it mean no (zero mean low

high and low), so custom write this function. What it will zero mean? This zero mean no fire it will send a zero signal to do port number one it will send, which there is no fire. So this zero mean no fire and it will send a one(one mean there is no fire), no fire is detected, so if one it will send a one signal one(one there is fire), so mean if it is detected one from detailed port number zero from detail port number zero if this detected one, fire is detected and then it will act accordingly the sprinkle with get on and it will start sprinkling the water.

```
1 from gpio import *
2 from time import *
3
4 def handleSensorData():
5     value = digitalRead(0)
6     if value == 0:
7         customWrite(1, '0')
8     else:
9         customWrite(1, '1')
10
11
12 def main():
13     add_event_detect(0, handleSensorData)
14
15     while True:
16         delay(1000)
17
18 if __name__ == "__main__":
19     main()
20
21
```

Figure2. program here we need to selected the python

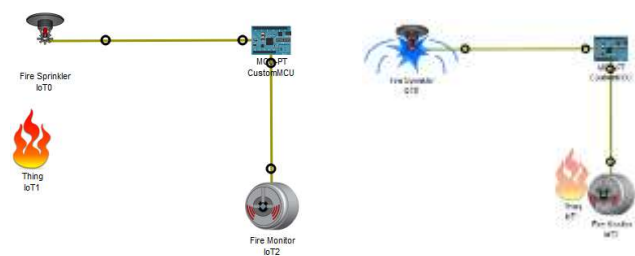


Figure. a

Figure. b

Figure. A	Figure. B
customWrite(1, '0')	customWrite(1, '1')
No Fire	Fire

Create an Thing (Heating Element) and in the script set a fire property with `setDeviceProperty(getName(), 'IR', 900);` For fire function we need java code `setDeviceProperty(getName(), 'IR', 900);` which works with the `setDeviceProperty` function:

```
1 function setup()
2 {
3   setDeviceProperty(getName(), 'IR', 900);
4 }
```

2. Conclusion

The danger of fire always threatens a building complex, and if it is not detected in time, it may cause a danger to the lives of the residents and create a disaster. For this reason, the use of a fire protection system is very important and vital. For this purpose, the fire detection and notification system should be used to detect the presence of fire in the environment and provide the possibility of appropriate response. Due to the importance of this issue, the use of fire sensor or smoke detection has a long history and was used in large complexes before. By using the Internet of Things, the possibility of online and real-time monitoring of the status of smoke detection sensors is also provided. In this way, it is possible to be informed about the occurrence of fire and its exact location in every part of the building and inform the relief organizations about this issue. In addition, due to the constant monitoring of this service, in case of any errors and failures in the system, you can be informed about the problem in advance and take the necessary measures. Also, by using the Internet of Things, it is possible for the smoke detection sensor to be easily integrated with different fire extinguishing systems so that the fire extinguishing system is also activated if a fire is detected. On the other hand, the use of new technologies in Internet of Things services makes the use of fire detection and alarm systems economical in small complexes, and therefore it is expected that, unlike in the past, a wider range of people and businesses can guarantee the safety of their building units.

Reference

[1]- Gillis, Alexander (2021). "What is internet of things (IoT)?" . IOT Agenda. Retrieved 17 August 2021.

[2]- Z. S. Obanda, "Multi-sensor fire detection system using an," Arduino Uno microcontroller, vol. 10, no. 03, pp. 23–26, 2017